

Initialization

main.py

init DBR_Spectrometer object
(all scan parameters set here)

laser.enable()
laser.disable()

laser.scan()
mode = "manual"

DBR.py
_manual_setup()

DBR.py
glob existing DAC
Table

Command Line
Prompt Settings

Use Existing DAC Table:
DAC_Table.get_DAC_arrays()

*/DBR/DAC_Tables

init DAC_Table object

DAC_Table.generate_DAC_table()

generate_table.py
get_DAC_arrays(Default LUT)

interpolation_type =
linear / curve_fit

_curve_fit()

_linear_interpolate()

_write_DAC_table()

*/DBR/DAC_Tables

DBR.py
DAC_Table.get_DAC_arrays()

plot()

plot_choice =
True / False

*/DBR/Plots

LUT Generation

_send_packets(DAC_list)

manual =
True / False

Command Line
Prompt Sending Packets
(num_packets, delay)

_make_packets_list(DAC_list)

_laser_serial.write(packet)
_read_response()
(for each packet)

response = 0x01

log_voltage =
True / False

_read_voltage()

_write_voltage()

Exit Program

Communication Protocol

*/DBR/Voltage_Data